Newton's method.

1. 原始牛顿法：求方程的根：如：$x^3-2x-5=0$，$e^x+x=0$，$\cos x = x$　$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$

2. 最优化牛顿法：求解无约束优化问题．$x_{n+1}=x_n-\left[\nabla^2 f(x_n)\right]^{-1}\nabla f(x_n)$

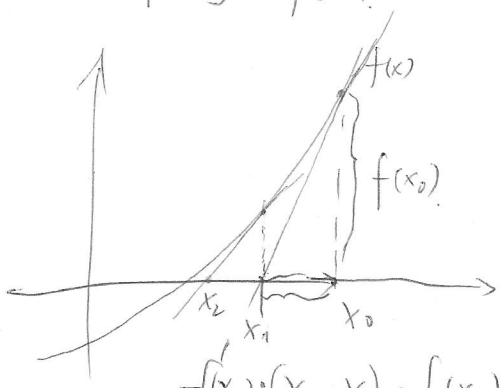
例子：$f(x)=x^2-2=0$

$$x_{n+1}=x_n-\frac{x_n^2-2}{2x_n}=\frac{x_n+\frac{2}{x_n}}{2}$$

$x_0=1$

$x_1=1.5$

$x_2=1.41667\ldots$



The goal of Newton's method for estimating a solution of an equation $f(x)=0$ is to produce a sequence of approximations that approach a solution.

$$f'(x_0)\cdot(x_0-x_1)=f(x_0)$$

$$\Rightarrow x_1=x_0-\frac{f(x_0)}{f'(x_0)}$$